Kinam Kim

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Research Interest

My research is on **world models for Physical AI**: making **video generative models** into simulators of the physical world [C1-C3, P1-P2], and turning those simulators into robot behavior [U1, C4-C6, O1]. On the generative side: sampling, conditioning and spatiotemporal consistency. On the robot side: latent reasoning inside **VLA** policies, reinforcement learning and zero-shot sim-to-real, and manipulation data generated without teleoperation.

Keywords: World Models, Video Generation and Diffusion Models, Vision-Language-Action (VLA) Models, Reinforcement Learning and Sim-to-Real

Honors and Service

- **Outstanding Paper Award**, Robotics: Science and Systems (RSS) 2026, for *FlashSAC* [C5]
- **Reviewer**, ICLR 2027, NeurIPS 2026, CoRL 2026

Education

KAIST, Graduate School of AI — Daejeon, South Korea
M.S./Ph.D. Integrated (Advisor: Prof. Jaegul Choo) — *Feb. 2025 - Present*
- GPA: 4.0 / 4.3

KAIST, School of Computing — Daejeon, South Korea
B.S. in Computer Science — *Mar. 2019 - Feb. 2024*
- GPA: 3.9 / 4.3

Work Experience

NVIDIA
Research Intern — *Jul. 2026 - Present*
- Latent-reasoning VLA with directly supervised reasoning states, under review [U1]

Microsoft Research Asia — Tokyo, Japan
Research Intern — *Feb. 2026 - Jul. 2026*
- Object-centric residual RL for VLA policies, zero-shot sim-to-real transfer (CoRL 2026) [C4]

KAIST DAVIAN Lab — Daejeon, South Korea
Undergraduate Research Intern — *Aug. 2023 - Mar. 2025*
- Sampling and fine-tuning methods for video diffusion models [C2-C3]

Publications

Under Review

[U1] **GistAct: Acting on Latent Answers in Vision-Language-Action Models**

Kinam Kim, Fu-En Yang, Min-Hung Chen, Zhiding Yu, Yu-Chiang Frank Wang, Chi-Pin Huang

Work done during an internship at NVIDIA

TL;DR: GistAct supervises a VLA's latent reasoning directly rather than distilling it from a text chain-of-thought teacher, ending that reasoning in a goal map over the scene. The action expert reads only the map, never the instruction, and reaches 67.8% success on RoboSPA spatial-24 against 33.9% for the strongest baseline.

World Models and Video Generation

[C1] **EgoX: Egocentric Video Generation from a Single Exocentric Video**

Taewoong Kang*, **Kinam Kim***, Dohyeon Kim*, Minho Park, Junha Hyung, Jaegul Choo

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026

TL;DR: EgoX generates egocentric video from a single exocentric video, carrying human appearance and motion through a large viewpoint change while keeping the scene spatially consistent.

[C2] **Temporal In-Context Fine-Tuning for Versatile Control of Video Diffusion Models**

Kinam Kim*, Junha Hyung*, Jaegul Choo

The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025

TL;DR: Temporal In-Context Fine-Tuning (TIC-FT) inserts noisy buffer frames to adapt a pretrained video diffusion model to new conditioning signals, using as few as 10 to 30 samples.

[C3] **Spatiotemporal Skip Guidance for Enhanced Video Diffusion Sampling**

Junha Hyung*, **Kinam Kim***, Susung Hong, Min-Jung Kim, Jaegul Choo

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025

Integrated into LTX-Video and Hugging Face Diffusers

TL;DR: Spatiotemporal Skip Guidance (STG) is a training-free sampling method that raises video diffusion quality by simulating a weak model through selective layer skipping, without the diversity loss of CFG.

[P1] **Cross-Frame Representation Alignment for Fine-Tuning Video Diffusion Models**

Sungwon Hwang*, Hyojin Jang*, **Kinam Kim**, Minho Park, Jaegul Choo

arXiv preprint, 2025

TL;DR: Cross-frame Representation Alignment (CREPA) regularizes video diffusion fine-tuning against neighboring frames, improving fidelity and cross-frame coherence under LoRA.

[P2] **Overcoming Camera-Facing Gaze Bias in EG3D Scene Generation**

Kinam Kim, Juhyun Lee, Youngdo Lee

arXiv preprint, 2023

TL;DR: Folding a pretrained gaze estimator into EG3D's training pipeline corrects the 3D GAN's camera-facing gaze bias on FFHQ without losing image quality.

Robot Learning

[C4] **Object-Centric Residual RL for Zero-Shot Sim-to-Real VLA Enhancement**
Kinam Kim, Namiko Saito, Heecheol Kim, Katsushi Ikeuchi, Jaegul Choo, Yasuyuki Matsushita
Conference on Robot Learning (CoRL), 2026
Work done during an internship at Microsoft Research Asia
TL;DR: An object-centric residual policy trained in simulation augments a base VLA and transfers zero-shot to the real world, raising success rates on tasks such as pick-and-place and drawer closing.

[C5] **FlashSAC: Fast and Stable Off-Policy Reinforcement Learning for High-Dimensional Robot Control**
Donghu Kim*, Youngdo Lee*, Minho Park, **Kinam Kim**, I Made Aswin Nahendra, Takuma Seno, Sehee Min, Daniel Palenicek, Florian Vogt, Danica Kragic, Jan Peters, Jaegul Choo, Hojoon Lee
Robotics: Science and Systems (RSS), 2026
Outstanding Paper Award
TL;DR: FlashSAC cuts Soft Actor-Critic's gradient updates and compensates with larger models and higher throughput, with norm bounds that keep the critic stable. It beats PPO and strong off-policy baselines on over 60 tasks across 10 simulators, and trains sim-to-real humanoid locomotion in minutes rather than hours.

[C6] **ACG: Action Coherence Guidance for Flow-based VLA models**
Minho Park*, **Kinam Kim***, Junha Hyung, Hyojin Jang, Hoiyeong Jin, Jooyeol Yun, Hojoon Lee, Jaegul Choo
IEEE International Conference on Robotics and Automation (ICRA), 2026
TL;DR: Action Coherence Guidance (ACG) is a training-free test-time method that removes noise-induced jerks and jitter in VLA policies, raising success rates on RoboCasa, DexMimicGen and real-world SO-101.

[O1] **Demonstration-Free Dexterous Manipulation Data at Scale**
Kinam Kim, Jaegul Choo
Ongoing research project, 2026
TL;DR: We generate bimanual dexterous manipulation data without a single human demonstration, on contact-rich tasks such as Rubik's cube turning, carabiner clipping and bolt fastening. A coding agent composes trajectory optimization, motion planning and scripted control into a task policy and revises its own code from simulation feedback.

Skills

- **Languages and frameworks:** Python, PyTorch, multi-GPU and multi-node distributed training
- **Robot learning:** VLA models (GR00T, $\pi_{0.5}$, SmolVLA), flow-matching policies, residual and off-policy RL
- **Simulation and real robots:** MuJoCo, RoboCasa, DexMimicGen, SO-101, sim-to-real transfer
- **Generative models:** video diffusion fine-tuning (Wan2.1, CogVideoX, Mochi), LoRA

Reference

- **Jaegul Choo**, Ph.D. Advisor, KAIST AI • jchoo@kaist.ac.kr
- **Yu-Chiang Frank Wang**, Internship Manager; Research Director, NVIDIA • ycwang@ntu.edu.tw
- **Yasuyuki Matsushita**, Internship Manager; Senior Director, MSRA Tokyo • yasumat@ist.osaka-u.ac.jp